

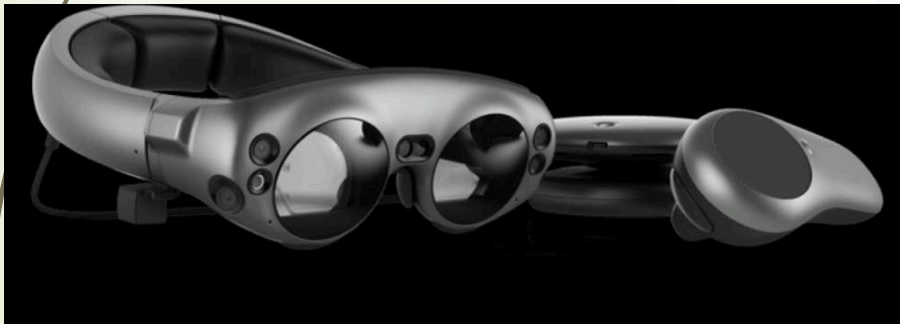
See-Through HMDs for Augmented Reality



Microsoft HoloLens



EPSON BT-30

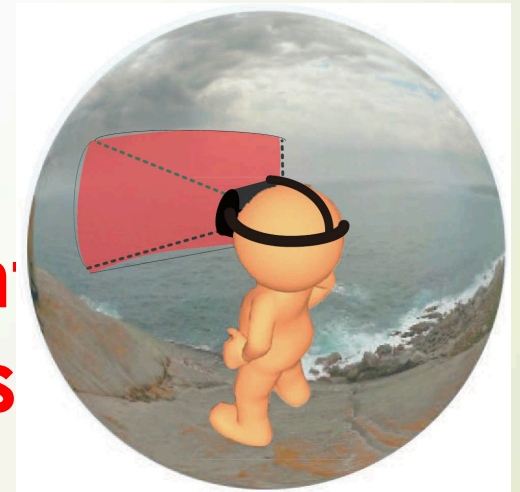


Magic Leap One



360 Videos Are Also Known as...

- Omnidirectional videos
- Spherical videos
- Panoramic videos
- Zoomable videos
- ...
- **Q: How is it different from regular videos**

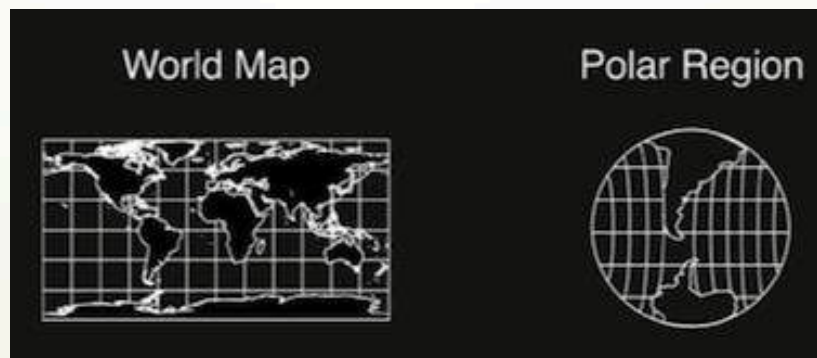


Agenda

- Cameras and Displays
- Projection Models
- Tiled Streaming (PA1)
- Edge Rendering
- Fixation Prediction
- Other Challenges

Projecting 360 Videos to 2D Rectangular Videos

- **Q: Why do we need to project 360 videos to 2D rectangle?**
- But, there exists no distortion free projections! ← that's why there are many proposals...



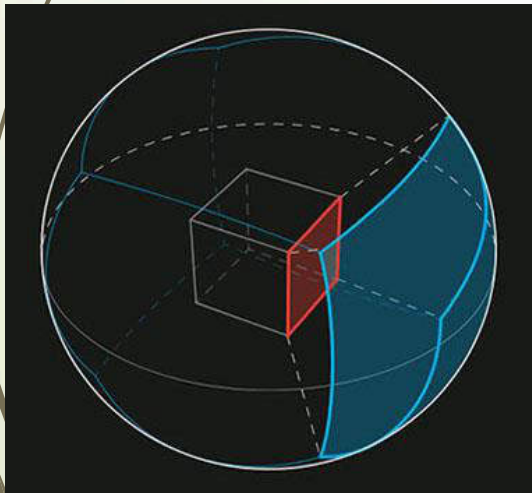
Equirectangular Projection

Other Projection Models...

- Cubic
- Pyramid
- Offset Cubic
-

Most variations vary the pixel density In different areas....

Q: What for?



Equirectangular is Still Widely Used for it's Simplicity

- Some optimization has been done in streaming service provider, e.g., at some one point, Facebook adopts **offset cubic** projections with **22** different orientations [Zhou et al., MMsys'17]
- **Q: Why Facebook needs to duplicate 360 videos in different orientations?**

